

Marcus Ma

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EDUCATION

University of Southern California

PhD in Computer Science (SAIL, Advisor: Shrikanth Narayanan)

August 2024 – Present

- 4.0 GPA; Courses taken: PSYC 626 Text as Data, CSCI 670 Advanced Analysis of Algorithms, CSCI 699 History of Language and Computing

Georgia Institute of Technology

MS in Computer Science, Machine Learning Specialization (Advisor: Wei Xu)

August 2023 – May 2024

- 4.0 GPA; Relevant Coursework: Game AI, Advanced Natural Language Processing, Theory of ML
- Master's Thesis: leveraging novel NLP techniques and LLMs for cross-genre authorship attribution and verification; creating a new SOTA domain-adaptive attribution model for multi-genre documents

BS in Computer Science, Intelligence and Theory Concentration

August 2020 – May 2023

- 4.0 GPA (graduated summa cum laude, Dean's List every semester)

HONORS & AWARDS

- Andrew and Erna Viterbi Graduate Student Fellowship, USC (2024-2026)
- DAAD RISE Research Scholarship, German Academic Exchange Service (2022)

SELECTED PUBLICATIONS

- Ma, M.**, Narayanan, S. Paired Multimodal Scaling Laws. Under Review at ICLR 2027.
- Ma, M.**, Narayanan, S. Intelligence Requires Grounding Beyond Embodiment. Under Review at AAAI 2027.
- Ma, M.**, Zhou, E., Ludwig, L., Hörath, J., Winkler, C., Avramidis, K., Feng, T., Toth, G., Bothe, A., Narayanan, S. Smiling Regulates Emotion During Traumatic Recollection. Proceedings of the 28th ACM International Conference on Multimodal Interaction (ICMI), 2026.
- Ma, M.**, Prescott, J., Zhou, E., Feng, T., Avramidis, K., Toth, G., Narayanan, S. Encoding Emotion Through Self-Supervised Eye Movement Reconstruction. Proceedings of ICASSP 2026, May 2026.
- Avramidis, K., Jeong, W., Kommineni, A., Kadiri, S., **Ma, M.**, McDaniel, C., Hughes, M., McGee, T., Kaiser, E., Byrd, D., Habibi, A., Cahn, B., Blank, I., Lerman, K., Medani, T., Leahy, R., Narayanan, S. Deep Learning Characterizes Depression and Suicidal Ideation in Young Adults from Eye Movements. npj Digital Medicine, 9(1), Mar 2026.
- Ma, M.**, Johnson, C., Bridges, N., Trager, J., Chochlakis, G., Narayanan, S. Authors Should Label Their Own Documents. arXiv preprint arXiv:2512.12976, Dec 2025.
- Ma, M.**, Chochlakis, G., Pandiyan, N. M., Thomason, J., Narayanan, S. Large Language Models Do Multi-Label Classification Differently. Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, Nov 2025.
- Chochlakis, G., Wu, P., Bedi, T., **Ma, M.**, Lerman, K., Narayanan, S. Humans Hallucinate Too: Language Models Identify and Correct Subjective Annotation Errors With Label-in-a-Haystack Prompts. Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, Nov 2025.
- Hall, K., **Ma, M.**, Zhang, X., Das Swain, V., Kim, Jennifer. Understanding Human-AI Misalignment in LLM-Based Job-Seeking Support for Neurodivergent Users. Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility, Oct 2025.
- Shi, X., Zhang, Y., Lu, Y., **Ma, M.**, Feng, T., Toutios, A., Hsu, H., Goldstein, L., Narayanan, S. 75-Speaker Annot-16: A benchmark dataset for speech articulatory rt-MRI annotation with articulator contours and phonetic alignment. Interspeech 2025.
- Ma, M.**, Le, D., Kang, J., Dou, Y., Cadigan, J., Freitag, D., Ritter, A., Xu, W. CROSSNEWS: A Cross-Genre Authorship Verification and Attribution Benchmark. Proceedings of the AAAI Conference on Artificial Intelligence, 39(23), 2025.

- **Ma, M.**, Glass, L. and Stanford, J. Introducing Bed Word: A New Automated Speech Recognition Tool for Sociolinguistic Interview Transcription. *Linguistics Vanguard*, 2024.
- **Ma, M.**, Kim, C., Hall, K., Kim, J. It Takes Two to Avoid Pregnancy: Addressing Conflicting Perceptions of Birth Control Pill Responsibility in Romantic Relationships. *Proc. ACM Human-Computer Interaction*. 7, CSCW2, Article 282 (October 2023).

ONGOING RESEARCH PROJECTS

Environmental Factors of General Intelligence

2025 – Present

USC SAIL (Narayanan)

- Argued in a position paper that intelligence requires grounding (consistent, causal assignment of meaning to symbols through interaction with an environment) but not embodiment (Under Review at AAAI 2027)
- Introduced a four-channel multimodal scaling law based on partial information decomposition, showing that only paired data reduces synergistic loss and that synergy emerges only past a critical number of pairs (Under Review at ICLR 2027)

Nonverbal Emotion and Cognitive State Analysis

2025 – Present

USC SAIL (Narayanan) + USC Shoah Foundation

- Pretrained a self-supervised gaze model on thousands of hours of unlabeled, low-resolution Holocaust survivor testimonies to predict speech-derived valence, arousal, and dominance and behaviors such as laughing and crying (ICASSP 2026)
- Built an automatic smile detector (85% F1) and aligned smiles from 978 survivors with narrative and multimodal valence features, finding negative-affect smiles precede more positive valence trajectories (ICMI 2026)

LLM Subjectivity Bias (Published in EMNLP 2025)

2024 – 2025

USC SAIL (Narayanan)

- Investigating prior biases of LLMs by investigating how often they attend to incorrect information directly provided during in-context learning versus relying on their pretrained internal representations
- Explaining multi-label, multi-class probability distributions through an aggregation of binary choice prompts rather than a single multi-label prompt distribution which inaccurately reflects model beliefs

Automatic Vocal Tract Segmentation (Published in Interspeech 2025)

2024 – 2025

USC SAIL (Narayanan) + SPAN

- Created a fully-automated pipeline for vocal tract area segmentation using a combination of hand-annotations and finetuned SAM-2 object segmentation models, increasing accuracy compared to off-the-shelf SAM2 by 70% and other vocal tract segmentation models by 15%

PREVIOUS RESEARCH LABS

NLP-X Lab (Natural Language Processing) | Advisors: Wei Xu and Alan Ritter

Cross-Domain Authorship Analysis (Published in AAAI 2025)

January 2022 – August 2024

- Created the largest public cross-genre authorship dataset across hundreds of different news sites using a combination of hand-picked gold labels and thousands of silver-labelled authors determined from statistical models of social media network graphs and string linkage methods
- Leveraged this dataset to create a domain-adaptable authorship attribution model trained via contrastive learning on negative-mined paraphrases that reaches SOTA performance on various authorship tasks

Co-Well Lab (Human-Computer Interaction) | Advisor: Jennifer Gahee Kim

Personalized LLMs for Neurodivergent Jobseekers (Published in ASSETS 2025)

Mar 2023 – Aug 2024

- Analyzed 300+ chat logs and interviewed 15 neurodivergent participants, finding that a GPT-4 career chatbot misrepresented users' skills and imposed neurotypical language and expectations

- Developed an app prototype designed to split the responsibility of taking pills more equitably between partners and interviewed nine couples to investigate the intersection of contraception responsibility and technology, findings published in CSCW 2023

SELECTED WORK HISTORY

Echo Group (Valuation: \$25M)

Chief Technology Officer & Co-Founder

2025 – Present

- Lead a team of 3 full-time SWE's and a dozen contractors to build a proprietary chatbot system with 100k US users and 10M+ chats sent that we use to understand population-level trends
- Develop and collect real-time, author-labeled preferences directly from human sources rather than LLM-based simulation better, faster, and more accurate than any other alternative on the market
- Provide insights to United, NextDoor, Waymo, GSK, J&J, Royal Caribbean, and several ex-Congressmen and political consultants
- Developed a backend that has processed 600M+ SQL database calls and 90M+ internal API calls across ~30 production services and ~60 code repositories

Mercor

LLM Software Engineer: Team Lead | Part-Time

Jan 2025 – July 2025

- Leading team of 150+ professional software engineers in a large-scale LLM human evaluation project for several large AI companies
- Overseeing generation and review of \$500,000+ worth of human-annotated LLM responses for cutting edge NLP research and production model Reinforcement Learning from Human Feedback (RLHF)
- Oversaw collection of expert-annotated math and coding documents used by frontier labs as verifiable rewards for training models such as Gemini, Claude, and GPT Codex

Amazon Web Services

Software Developer Intern: Optical Networking Team | Full-Time Position Offered May 2023 – August 2023

- Spearheaded the transition of optical device networking communication from internal Amazon servers to the more scalable and maintained Native AWS environment, saving ~\$10,000 of yearly server operation costs
- Created a DNS device hostname resolution script that was deployed to hundreds of existing optical devices in the AWS fleet, improving operational efficiency over previous method by ~40%

Software Developer Intern: Postgres SQL Team | Full-Time Position Offered

May 2022 – August 2022

- Worked in Postgres source code to implement an optimization on the SQL Join command that used semi-join hash filtering to remove unnecessary table rows early in the execution tree
- Improved the speed of certain SQL join commands by up to 36% and integrated filter decisions for the merge join strategy into Postgres query planner, improving AWS Postgres' overall TPC-H database benchmark performance by ~5%

TEACHING EXPERIENCE

Graduate Teaching Assistant for CS 4650: Natural Language Processing Jan 2023 – May 2024

- Assist in leading classes of 75+ students through NLP, covering topics from encoders to present-day LLMs

Tutoring and Academic Services: One-on-one Tutor, PLUS Leader

Aug 2021 – Dec 2022

- Prepared and taught ~30 weekly PLUS Sessions (~1200 student cumulative attendance) for Data Structures